

Tactile Graphics Exploration Studies Using Fingertip Tracking Based On Colour Markers Detection For Visually Impaired People

Muhammad Ikmal Hakim Shamsul Bahrin
Department of Mechatronics Engineering,
Kulliyah of Engineering
International Islamic University Malaysia
Kuala Lumpur, Malaysia
muhammad.ikmalhakim@gmail.com

Hazlina Md Yusof
Department of Mechatronics Engineering,
Kulliyah of Engineering
International Islamic University Malaysia
Kuala Lumpur, Malaysia
myhazlina@iiu.edu.my

Shahrul Na'im Sidek
Department of Mechatronics Engineering,
Kulliyah of Engineering
International Islamic University Malaysia
Kuala Lumpur, Malaysia
snaim@iiu.edu.my

Abstract— It is very difficult for people with visual impairment and blindness to access visual information. Thus, tactile graphics became one of the alternatives to help them access maps, graphs, painting, illustration and other graphical materials. However, it is challenging to use tactile graphics. The blind and visually impaired people need to have good exploratory procedures and strategies to gather information tactually. In psychology, several studies have been conducted to clarify how people with visual impairment and blindness perform tactile graphics exploration. But the information is still insufficient, and the studies conducted usually relies on video observation. Therefore, to provide more insight on how people with visual impairment and blindness explore tactile graphics, a fingertip tracking system using Logitech Webcam and MATLAB software has been developed. The method of tracking the fingertips is by using colour markers detection. With the system, the position of the fingertips can be recorded and used for analyzing the tactile graphics exploration activities. It is aimed that this system can be a tool to visualize the cognitive process of people visual impairment and blindness explore tactile graphics materials.

Keywords—Visual Impairment; Blindness; Tactile Graphics / Tactile Diagram; Tactile Cognition; Tactual Perception; Tactual Exploration; Hand Tracking System; Finger Tracking System

I. INTRODUCTION

Tactile graphics are one of the alternatives for people with visual impairment and blindness to access visual information such as pictures, maps, diagrams, graphs, symbols, and other types of non-textual information. They are important for a variety of purposes such as education, navigation and entertainment. For example, in blind schools, tactile graphics have been used for STEM (Science, Technology, Engineering and Mathematics) subjects [1], illustrations in children's book [2] and learning maps and geography [3]. To obtain information from the tactile graphics, visually impaired and blind people need to use their hands and fingers, to touch and feel the raised surfaces of the figures displayed. They will need time to examine, synthesize and understand the tactile information [5].

Tactile graphic exploration is challenging, and it is a complex task. Every blind and visually impaired people have different level of performance during the exploration due to different levels of ability and different reaction towards tactile information [6][7]. When identifying figures in a tactile format, they need instructions and guidance with physical contact in order to organize tactile stimuli to construct a mental representation of the image [4][8]. Not

only that, they need to be trained with exploratory procedures and strategies to extract information from a haptic scene [9]. Although, there are techniques and methods to help them explore tactile information still, it is unsure how do people with visual impairment and blindness understand graphical information through touch sensing. The information is still limited and need to be investigated further [10][11][12]. By investigating the way they explore tactile graphics, the data gathered will be useful for designing effective representation of the tactile graphics materials, providing guidelines and teaching techniques to explore graphical information and designing of accessible interface on touchscreens [13][14][15][16][17].

Therefore, this paper is presenting a prototype that can be used to investigate how people with visual impairment and blindness explore tactile graphics. It is a computer vision system that performs fingertips tracking based on colour markers detection method. When people with visual impairment and blindness are exploring tactile graphics materials, the system can detect and record the position of the colour markers that are placed on his or her fingertips. However, the system can be used to track only five fingers of a single hand. Once the positions of the fingertips have been recorded, further analysis on their exploring behaviour and strategies need to be performed. The recorded fingertips' positions will be plotted to visualize the fingers moving directions and the pattern of exploring the tactile graphics materials. From the result, it could be used to highlight how people with visual impairment and blindness understand and acquire graphical information through touch sensing.

In future, the system will be tested on people with visual impairment and blindness and they will be given task to read tactile graphics materials that have been obtained from KL Braille Resources located at Sentul, Kuala Lumpur. There are textured and raised line tactile graphics that are showing simple and complex diagrams. The structure of the paper is organized as follows: section II briefly reviews the background on visual impairment, tactile graphics design and production and challenges in reading tactile graphics materials. Section III presents related and previous works performed by other researchers. Section IV describes about the system design, experimental setup and results. Lastly, section V concludes the paper and highlights ongoing challenges and future directions.

II. BACKGROUND

A. Visual Impairment

Visually impaired people refers to people that had functional loss of vision at a certain level of visual impairment [18]. Visual impairment is related to the word blindness which is ranging from legal blindness to total blindness as shown in Table 1. According to World Health Organization, there are about 285 million of visually impaired people worldwide [19]. In Malaysia, there are about 50,000 visually impaired people who have registered with Malaysian Social Welfare Department and there are still 50 000 left unregistered [20]. In 2018, Health Minister Dr Dzukefly Ahmad mentioned that, there are at least 1.2% or 63,000 Malaysians who had been diagnosed as blind and people with limited vision are some 350, 000 people or 5.4% according to National Eye Survey in 2014 [21] The causes of the blindness or limited vision are because of cataract, diabetes and glaucoma. From the survey, the eye disease is expected to increase in future because of ageing population and increase in diabetic cases in Malaysia.

TABLE I. CLASSIFICATIONS OF THE LEVEL OF VISUAL IMPAIRMENT [18]

Level	Ratio ^a
Moderate low vision	20/80
Severe low vision	20/200
Profound low vision	20/500
Near blindness	20/1250
Total blindness	No light perception

^a the ratio here means that, a low vision person needs to be at 20 feet arc to see an object, while a normal person can see clearly an object in front at 200 feet only

B. Accessing visual information via Tactile Graphics

Tactile graphics or tactile diagrams are raised figures of visual information such as illustration, graphs, maps, diagrams and other type of drawings [22]. The main purpose of tactile graphics is to deliver a concept of visual picture or non-textual information in tactile format [2][6]. To obtain visual information like sighted people looking at pictures, blind or visually impaired people need to use tactile graphics materials and must utilize their touch senses to read and explore every information on them. However, there are limitations to interpret information on tactile graphics with tactual perception. The view of tactile graphics is limited to the size of a person's fingertips. Thus, a person's tactual perception has a lower resolution and smaller field of view which result in lower efficiency to acquire information [23]. For acquiring information from tactile graphics, the tactile graphics readers must be developed with tactual-kinaesthetic skills and need to use a set of exploratory procedures. According to Barraga and Erin [24], tactual-kinaesthetic development is said to begin with an awareness of and attention to the differences in size, shape, texture, and weight of objects. While, exploratory procedures mean hand movement patterns that can be seen when a person is accessing object properties using their haptic sense [25][26].

C. Design and Production of Tactile Graphics

Tactile graphics can be made into two-dimensional tactile graphics (2D) as shown in Fig. 1 and three-dimensional (3D) dimensional tactile graphics as shown in Fig. 2. They need to be prepared by a sighted individual who is highly skilled in designing and crafting the diagram. Not only that, the designer needs to ensure that the diagram can be understood by people with visual impairment and blindness. However, the process of making tactile graphics materials can be a labour-intensive and time consuming. There are many ways on how tactile graphics can be made. Table 2 shows a summary on techniques and technologies available for producing tactile graphics according to [6] [27] [28] and in Fig. 3 is showing process of making tactile graphics.

TABLE II. TECHNIQUES AND TECHNOLOGIES AVAILABLE FOR PRODUCING TACTILE GRAPHICS [6][27][28]

Method	How it is made?
Collage Method	A variety of material are glued to make a tactile image that may consist of lines, texture and symbols. The materials used can be a sandpaper, textured cloth, string, wire, pieces of tagboard etc. A master copy will be created and then it is used for thermoformed copies.
Embossing Foil	A master copy is made by embossing aluminum foil into labels, symbols and textured surface. A special kit must be used for embossing.
Capsule Paper	A popular method to create tactile graphics quick and easy. It is known as swell graphics, puff pictures and toaster graphics. Tactile images will be printed on a microcapsule paper after going through several processes like infrared light exposure and heating in a special machine
Paper Embossers	It uses braille embossers to produce tactile graphics. Tiger embossers manufactured by ViewPlus Technologies capable to produce dots of varying height and space for the tactile images. As a result, tactile graphics are produced in a form of line diagram.
Thermoforming	A method used to copy collage and embossed foil masters. It is a simple process where the master copy will be placed on a plate and a sheet of plastic called Brailon is placed on top. Then, heat is applied using thermoform machine to create a copy of tactile graphics

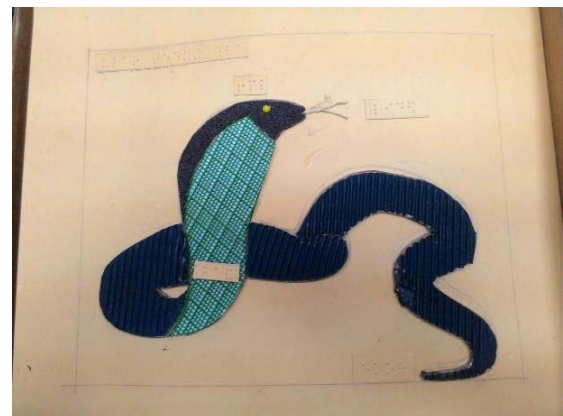


Fig. 1. An example of original 2D tactile graphics showing a snake diagram. This picture was taken during a visit to KL Braille Resources (27 Oct 2016)

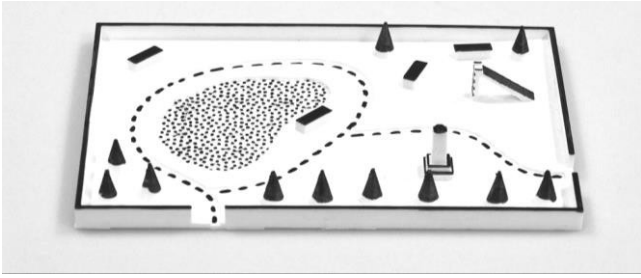


Fig. 2. A 3D printed map of a park from [29]



Fig. 3. A KL Braille Resources worker was using a thermoforming machine to create an emboss image from a capsule paper (27 Oct 2016).

III. RELATED WORKS

There are many approaches that have been done to study how people with visual impairment and blindness explore and understand tactile graphics. Postma et al. [30] have studied on haptic spatial-configuration learning and resulting memory traced between, early blind, late blind and blindfolded sighted individuals. They found out that, blind groups were much faster to match ten 3D shapes haptically. Mazella, et al. [31] have conducted a test to assess the perceptual-motor and cognitive skills involved in the processing of 2D tactile materials for children and adolescents with and without visual impairment. They created a new haptic test battery to measure scanning skills, haptic discrimination skills, spatial understanding, tactile short-term memory and tactile picture comprehension. Picard, Albaret and Mazella [32] have conducted a research on haptic identification of raised-line drawings between visually impaired and sighted children. They found out that, visually impaired children perform better compared to sighted children because they have superior exploration skills and they benefited semantic cueing on the tactile graphics.

There are also studies that have been conducted using computer vision system. Cheng et al. [17] have created an experimental technique to examine the cognitive processes involved in tactile graphics perception. They used Kinect camera to extract hand features of a blind and visually impaired people when exploring tactile graphics. Brock et al [14] have created a prototype using Microsoft Kinect camera and a multitouch table to analyse visually impaired people exploration strategies on tactile map. Bardot et al. [16] have used a touch screen and a camera to track hands and fingers movements during tactile exploration of raised-line

diagrams. Murugavel [12] has developed an automated finger tracking system that can track fingers movement from a captured video. The system was used to understand how people with visual impairment touch and feel tactile images Murai, Tatsumi and Miyakawa [11] have developed a system that can detect fingers' position during touch sensing of visually impaired people. They wanted to how visually impaired people recognize figures and shapes through analysing and evaluating fingers movement.

IV. SYSTEM DESIGN, EXPERIMENTAL SETUP AND RESULTS

With some references from the previous works and researches, a computer vision system has been developed for the investigation on the tactual behavior of visually impaired people when reading tactile graphics. The aim of the system is to monitor and track the position of fingers during tactile graphics reading in real time. To track the fingers, the system will detect colour markers that are placed on fingernails. However, only five fingers of a single hand can be tracked by the system. Once the fingers' movements have been recorded, the system can also be used to plot the positions of the fingers and visualize the path during tactile graphics reading. Hence, this section will discuss further on system design, experimental setup and results.

A. Logitech Webcam

Logitech webcam is one of the devices used in the research of computer vision. Usually Logitech webcam is being used for streaming, video recording and video conferencing. Here, a Logitech C922 as shown in Fig. 4 webcam model have been chosen to develop our computer vision system via MATLAB. This model can be used to stream at full 1080p at 30 frame per second or hyper fastHD 720p at 60 frame per second. The purpose of using this model are because of its high definitions view with fast frame rate, autofocus capability and light correction feature that still produces sharp high definition images even if the surrounding is dim.



Fig. 4. Logitech C922 webcam model

B. Building a fingertip tracking system using MATLAB

To develop the fingertips tracking system, MATLAB software has been used. It has user friendly prebuilt functions and Image Processing Toolbox to perform image processing techniques on images and videos. Other than that, a graphical user interface (GUI) has been designed using MATLAB as shown in Fig 5. The purpose of designing the system's GUI is to smoothen the process of conducting experiment such as recording video and storing the fingertips' coordinates of subjects. There are five phases of the system that have been described and shown in Fig 5, which are (i) storing experiment's information (ii) camera setup and preview (iii) fingertips tracking (iv) saving captured data and video and (v) visualization of fingertips path after tactile graphics

reading. For fingertips tracking algorithm, it comprises of the following methods; (i) image acquisition, (ii) locating and rectifying a tactile graphic diagram (iii) background subtraction, (iv) colour markers detection and segmentation, (v) morphological operations and (vi) fingertips tracking.

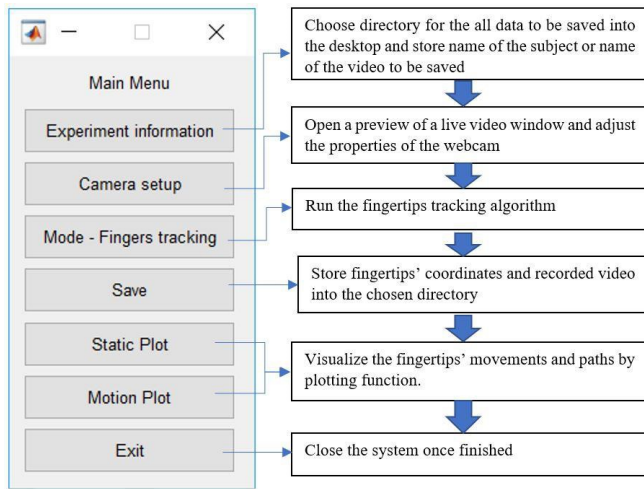


Fig. 5. System GUI and its description

C. Experimental setup

To conduct the experiment using the system, a room or a space with steady lighting conditions need to be chosen. This is to ensure constant illumination during the experiment, and the tracking system can run smoothly with good detection accuracy. Next, is to place the Logitech C922 webcam mounted on a fixed stand and the tactile graphics diagrams at a fixed position as shown in Fig. 6. When pointing the webcam towards tactile graphics diagram, make sure that the field of view of the webcam covers the whole area of the tactile graphics diagrams and there are no external objects on the reading area. The next procedure is to attach colour markers on fingertips of the subject. There are five colours that can be detected by the system and need to be placed as shown in Fig. 7. Once the setup is finished, the system needs to be operated in sequence according to the flowchart shown in Fig 5.

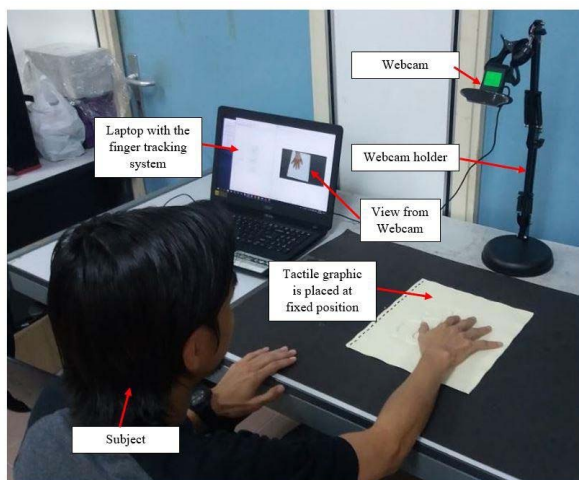


Fig. 6. View of the experimental setup of with a subject. The tactile graphic needs to be placed at a fixed position and the subject can only use one hand to read the tactile graphic.

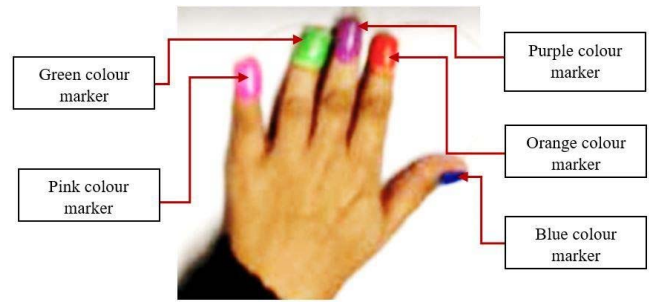


Fig. 7. Five different colour markers are placed on subject's hand

D. Results

This section presents the results obtained based on the algorithm used for the fingertips tracing system as explained in Section IV (B). In the first stage of the algorithm, an image data will be acquired from the webcam in 432 x 240 resolution as shown in Fig. 8. Based on the view captured, the tactile graphic's position is slanted. Thus, the system will locate and rectify the tactile graphic's position as shown in Fig. 9. The process has been done using perspective transformation method based on [33]. The next stage of the algorithm is the background subtraction which is shown in Fig. 10. The idea of the background removal is simply by subtracting the subsequent image data which has the hand placed on top of the tactile graphics and the background image data which is the rectified position of tactile graphic as in Fig. 9.



Fig. 8. The result of the image acquired from the webcam's perspective



Fig. 9. The result of located and rectified image data using perspective transformation method

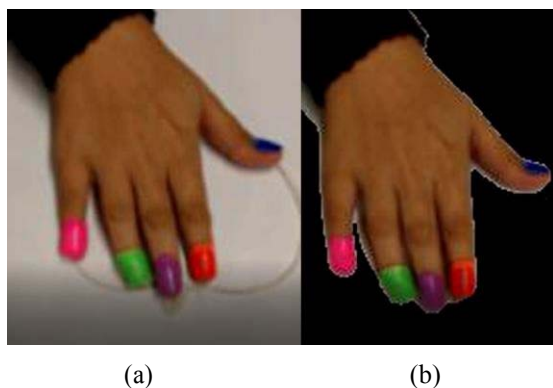


Fig. 10. Background subtraction is performed once the hand is placed on the tactile graphic. The left side (a) is the image data with background and the right side (b) is the result after the background has been removed.

Next is the colour markers detection and segmentation stage. The idea behind the process, is to use MATLAB's Color Thresholder App to generate five color markers' functions containing MATLAB codes. The codes will first perform color space conversion to the image data in Fig. 10 (b) from RGB color space to HSV color space and then create mask image based on every colour threshold value. As shown in Fig. 11, every colour marker has been segmented from the hand and the results are mask images in RGB color space. Then, morphological operations have been applied to binary mask image of every colour marker using sequence of dilation and erosion. The purpose is to obtain a smooth, closed and complete blob of colour markers in binary mask image as shown in Fig. 12. In the final stage of the algorithm, the movements of every colour marker will be tracked and recorded in form of x and y coordinate. To obtain the coordinates, the centroid of the blob in Fig. 12 has been extracted using 'regionprops' function in MATLAB. Hence, the system can visualize the movements, or the path taken by the fingertips by plotting the coordinates recorded. There are two ways on how the visualization process are done by the system which are (i) coordinates plotting in real time and (ii) using the 'Static Plot' and 'Motion plot' features for replotting the coordinates after any experiment conducted. Fig. 13 shows the results of the visualization process.

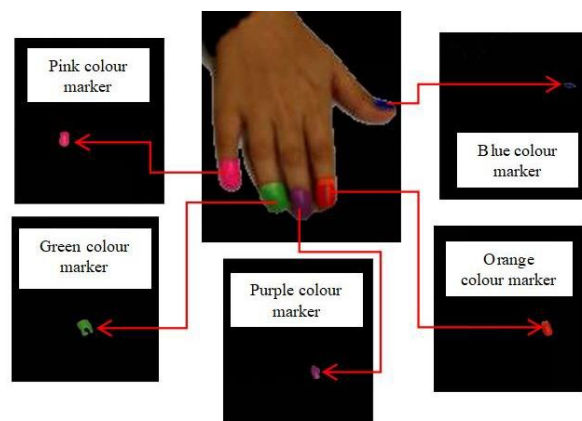


Fig. 11. The result of colour markers detection and segmentation using Color Thresholder App. The app creates mask image of every colour marker in RGB colour space

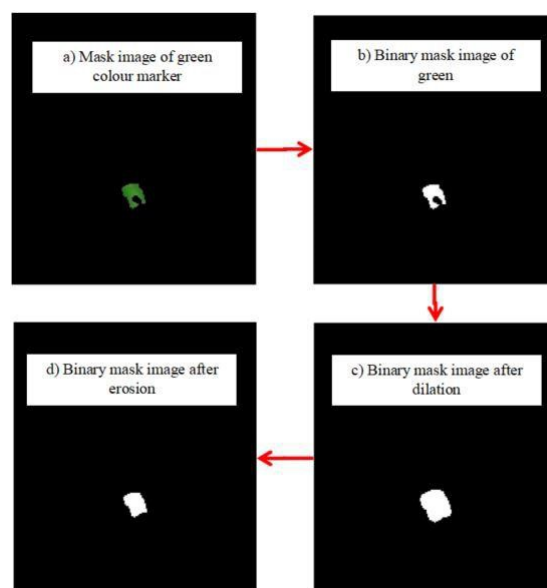


Fig. 12. Example results of morphological operations that have been applied to the binary mask image of green colour marker.

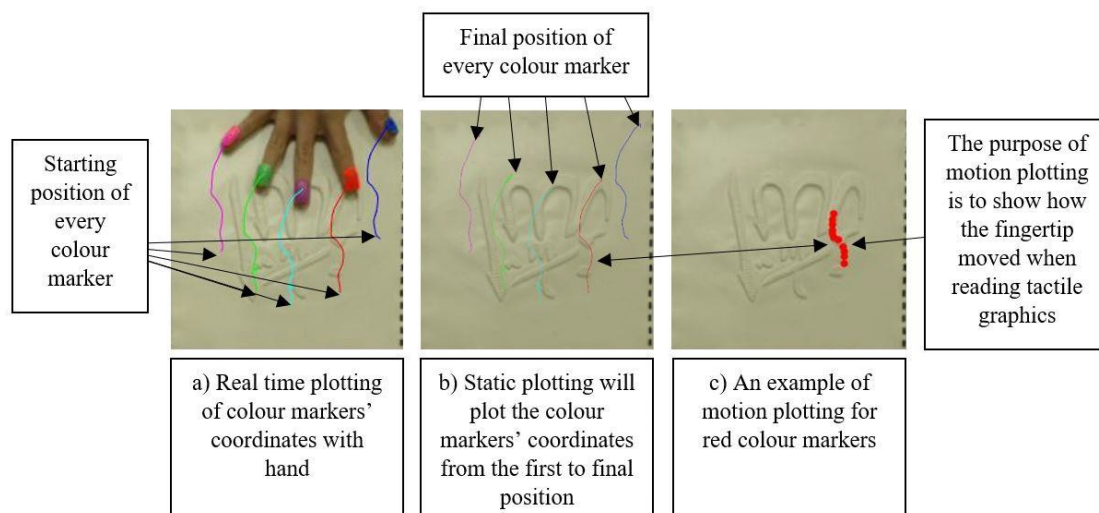


Fig. 13. Example results of the visualization process. Real time plotting in (a), static plotting in (b) and motion plotting in (c)

V. DISCUSSION & CONCLUSION

This paper has presented the fingertips tracking system based on colour markers detection method. The purpose of developing the system is for the investigation on how people with visual impairment and blindness explore tactile graphics materials. Based on the algorithm used, the system can detect and track the position of every colour marker placed on the fingertip from the beginning until the end. As shown in Fig. 13, the system can also visualize back the path and the movement performed by plotting every finger's coordinates. However, the system is currently limited to detect only five fingers of one single hand and need to be tested further under various conditions. In future, the system's algorithm needs to be improved to achieve more robust and better performance. Plus, it is hoped that the system can be used for the investigation mentioned and gather actual touch sensing data among the visually impaired and blind people when reading tactile graphics materials.

ACKNOWLEDGMENT

We would like to thank KL Braille Resources and Malaysian Association for the Blind (MAB) for all the supports and guidance given to conduct this research. This work was granted by FRGS/1/2018/TK04/UIAM/02/19.

REFERENCES

- [1] Smith, D.W. and Smothers, S.M., 2012. The role and characteristics of tactile graphics in secondary mathematics and science textbooks in braille. *Journal of Visual Impairment & Blindness*, 106(9), pp.543- 554.
- [2] Wright, S., 2008. Guide to Designing Tactile Illustrations for Children's Books. American Printing House for the Blind.
- [3] Ducasse, Julie, Anke M. Brock, and Christophe Jouffrais. "Accessible interactive maps for visually impaired users." In *Mobility of Visually Impaired People*, pp. 537-584. Springer, Cham, 2018.
- [4] David Dulin (2007) Effects of the use of raised line drawings on blind people's cognition, *European Journal of Special Needs Education*, 22:3, 341-353, DOI: 10.1080/08856250701430935
- [5] Downing, J.E. and Demchak, M., 1996. First steps: Determining individual abilities and how best to support students. Including students with severe and multiple disabilities in typical classrooms: Practical strategies for teachers, pp.35-61.
- [6] Pathstoliteracy.org. (n.d.). Tactile Graphics. [online] Available at: <http://www.pathstoliteracy.org/tactile-graphics> [Accessed 9 Nov. 2018].
- [7] Downing, J.E. and Chen, D., 2003. Using tactile strategies with students who are blind and have severe disabilities. *Teaching Exceptional Children*, 36(2), pp.56-61.
- [8] Dulin, D., & Hatwell, Y. (2006). The Effects of Visual Experience and Training in Raised-Line Materials on the Mental Spatial Imagery of Blind Persons. *Journal of Visual Impairment & Blindness*, 100(7), 414-424
- [9] McLinden, M. (2004). Haptic exploratory strategies and children who are blind and have additional disabilities. *Journal of Visual Impairment & Blindness*, 98(2), 99-115.
- [10] Murai, Y., Tatsumi, H., Sekita, I., & Miyakawa, M. (2016, October). Touch tracking analysis for graphics image acquisition by the visually impaired: Toward understanding graphical image creation by touch- sensing. In *2016 IEEE International Conference on Systems, Man, and Cybernetics (SMC)* (pp. 000367-000372). IEEE.
- [11] Murai, Y., Tatsumi, H., & Miyakawa, M. (2018, July). Recording of fingertip position on tactile picture by the visually impaired and analysis of tactile information. In *International Conference on Computers Helping People with Special Needs* (pp. 201-208). Springer, Cham.
- [12] Murugavel, A. (2014). Finger position data acquisition system for cross-modal tactile/visual cognition studies (Doctoral dissertation, San Diego State University).
- [13] Breidegard, B., Eriksson, Y., Fellenius, K., Holmqvist, K., Jönsson, B., & Strömquist, S. (2008). Enlightened: The art of finger reading. *Studia Linguistica*, 62(3), 249-260.
- [14] Brock, A., Lebaz, S., Oriola, B., Picard, D., Jouffrais, C., & Truillet, P. (2012, May). Kin'touch: understanding how visually impaired people explore tactile maps. In *CHI'12 Extended Abstracts on Human Factors in Computing Systems* (pp. 2471-2476). ACM.
- [15] Morash, V. S., Pensky, A. E. C., Tseng, S. T., & Miele, J. A. (2014). Effects of using multiple hands and fingers on haptic performance in individuals who are blind. *Perception*, 43(6), 569-588.
- [16] Bardot, S., Serrano, M., Oriola, B., & Jouffrais, C. (2017, May). Identifying how Visually Impaired People Explore Raised-line Diagrams to Improve the Design of Touch Interfaces. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (pp. 550-555). ACM.
- [17] Reginald Phillips Research Network for Improving Education for People. (2019, September). *Cognitive Science of Tactile Graphics : Research : The Reginald Phillips Research Network for Improving Education for People with Visual Impairment* (RM Phillips Research Network). Retrieved September 4, 2018, from <http://www.sussex.ac.uk/rmphillips/research/tactilegraph>.
- [18] (2002) Visual Impairment and Blindness. Independent Study Course Released: April 2002. Department of Veterans Affairs Employee Education System.
- [19] Pascolini, D. and Mariotti, S.P., 2012. Global estimates of visual impairment: 2010. *British Journal of Ophthalmology*, 96(5), pp.614- 618.
- [20] Ppobm.org.my. (2006). Pertubuhan Pembangunan Orang Buta Malaysia. [online] Available at: <http://ppobm.org.my/> [Accessed 15 Jun. 2019].
- [21] Benjamin, N. (2018). Health Ministry: 80 percent of eye diseases are preventable, treatable - Nation | The Star Online. [online] Thestar.com.my. Available at: <https://www.thestar.com.my/news/nation/2018/10/11/health-ministry-80-percent-of-eye-diseases-are-preventable-treatable/> [Accessed 19 Jun. 2019].
- [22] Way, T.P. and Barner, K.E., 1997. Automatic visual to tactile translation. i. human factors, access methods and image manipulation. *IEEE Transactions on rehabilitation engineering*, 5(1), pp.81-94.
- [23] Gupta, R., Balakrishnan, M. and Rao, P.V.M., 2017. Tactile diagrams for the visually impaired. *IEEE Potentials*, 36(1), pp.14-18.
- [24] Barraga, N. and Erin, J.N., 2001. Visual impairments and learning. Austin, TX: Pro-Ed.
- [25] Kalia, A., Hopkins, R., Jin, D., Yazzolino, L., Verma, S., Merabet, L., Phillips, F. and Sinha, P., 2014. Perception of tactile graphics: Embossings versus cutouts. *Multisensory research*, 27(2), pp.111-125.
- [26] Lederman, S. J. and Klatzky, R. L. (1987). Hand movements: A window into haptic object recognition, *Cogn. Psychol.* 19, 342-368.
- [27] Jaquiss, Robert Stuart. "An Introduction to Tactile Graphics." (2011).
- [28] Sheppard, L. and Aldrich, F.K., 2000. Tactile Graphics: A beginner's guide to graphics for visually impaired children. *Primary Science Review*, 65, pp.29-30.
- [29] Round Table. (2018). 3D printing for vision impaired students. [online] Available at: <http://printdisability.org/about-us/accessible-graphics/3d-printing/> [Accessed 15 Jun. 2019].
- [30] Postma, A., Zuidhoek, S., Noordzij, M.L. and Kappers, A.M., 2007. Differences between early-blind, late-blind, and blindfolded-sighted people in haptic spatial-configuration learning and resulting memory traces. *Perception*, 36(8), pp.1253-1265.
- [31] Mazella, A., Labardin, M., Mesnières, S., Albaret, J.M. and Picard, D., Construction of a 2D haptic test battery for use with children and adolescents with and without visual impairment. *Terra Haptica*, 4, pp.65-74.
- [32] Mazella, A., Albaret, J.M. and Picard, D., 2016. Haptic-2D: A new haptic test battery assessing the tactual abilities of sighted and visually impaired children and adolescents with two-dimensional raised materials. *Research in developmental disabilities*, 48, pp.103- 123.
- [33] Fusco, G., & Morash, V. S. (2015, October). The tactile graphics helper: providing audio clarification for tactile graphics using machine vision. In *Proceedings of the 17th International ACM SIGACCESS Conference on Computers & Accessibility* (pp. 97-106). ACM.